

Notes: Cooper and Haltiwanger (RES, 2006)

On the Nature of Capital Adjustment Costs

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1 Big picture

- **Question.** What is the nature of capital adjustment costs at the plant level? The paper asks whether investment dynamics are better explained by convex adjustment costs, fixed/non-convex costs, irreversibility, disruption costs, or a mixture of these frictions.
- **Core empirical fact.** Plant-level investment is highly non-normal: many observations show low or zero investment, some observations show large positive bursts, negative investment occurs but is less common, and investment responds only imperfectly to profitability shocks.
- **Data.** The paper uses plant-level data from the Longitudinal Research Database (LRD), focusing on moments of investment rates and profitability at the micro level.
- **Method.** The authors use indirect inference. They estimate structural parameters by choosing the model parameters that best reproduce a small set of moments summarizing the nonlinear relationship between investment and profitability.
- **Main result.** A model that mixes convex and non-convex adjustment costs fits the plant-level moments best. Pure quadratic/convex models or pure non-convex models miss important features of the data.
- **Economic takeaway.** Investment frictions are multidimensional. Plants face small smooth adjustment costs, but also important non-convex frictions such as disruption costs, irreversibility, and fixed adjustment costs. This implies that investment policy and aggregate investment dynamics cannot be well understood with a single quadratic adjustment-cost specification.

2 Incremental contribution of the paper

- **From reduced-form investment equations to structural adjustment costs.** Earlier investment work documents lumpy and intermittent plant-level investment. This paper estimates the underlying structural costs that can rationalize those moments.
- **A unified model of multiple frictions.** Instead of comparing only one friction at a time, the paper allows convex costs, fixed costs, irreversibility, and disruption costs to coexist.
- **Indirect inference for plant-level investment.** The estimation strategy targets moments that summarize inaction, investment bursts, serial correlation, and the investment-profitability relationship.
- **Model discrimination.** The paper shows which frictions are needed to match which empirical moments. Convex costs help with smooth responsiveness, while non-convex costs help match inaction and bursts.
- **Policy relevance.** Since adjustment costs shape how plants respond to tax credits, productivity shocks, and other policies, the estimated friction structure matters for macroeconomic policy evaluation.

3 Data and measurement

3.1 Investment and profitability moments

The key empirical object is the plant-level investment rate. The paper summarizes its distribution and its link to profitability using moments such as:

- average investment rate;
- fraction of observations with inaction;
- fraction of negative investment;
- positive and negative investment spike rates;
- serial correlation of investment rates;
- correlation between profitability shocks and investment.

The moments are designed to capture both **extensive-margin adjustment** (whether plants invest at all or make a burst) and **intensive-margin adjustment** (how much investment responds to profitability).

3.2 Empirical patterns

- Average investment is positive, but the distribution has a fat right tail.
- There is a meaningful inaction region, though not enough to be explained by a pure fixed-cost model alone.
- Negative investment occurs, so investment is not literally irreversible, but the asymmetry between positive and negative bursts indicates that irreversibility-like frictions matter.
- The serial correlation of investment is low, which is difficult for a pure quadratic model to match.

4 Model and adjustment-cost specifications

4.1 Plant problem

The plant chooses capital to maximize expected discounted value. A compact way to write the decision problem is:

$$V(A, K) = \max_{K'} \left\{ \Pi(A, K) - C(K', K, A) - p(K' - (1 - \delta)K) + \beta E[V(A', K') \mid A] \right\},$$

where A is profitability, K is capital, K' is next-period capital, $\Pi(A, K)$ is the profit function, and $C(\cdot)$ captures adjustment costs.

The profit function is commonly represented as:

$$\Pi(A, K) = AK^\theta.$$

4.2 Main adjustment-cost components

- **Convex cost** γ . Smooth adjustment cost that rises with the investment rate. This is the standard quadratic-style friction.
- **Fixed cost** F . A plant pays a fixed cost when it adjusts capital. This generates inaction and lumpy adjustment.
- **Irreversibility / resale wedge** p_s . Selling capital is costly; $p_s < 1$ means the resale price is below the purchase price.
- **Disruption cost** λ . Capital adjustment disrupts production; $\lambda < 1$ captures a loss in productive efficiency or output during adjustment.

4.3 Identification intuition

- **Inaction and bursts** discipline the non-convex components.
- **Serial correlation** disciplines whether adjustment is too smooth and persistent.
- **Correlation between investment and profitability** disciplines how sensitive capital is to productivity/profit shocks.
- **Positive versus negative spikes** helps distinguish fixed costs from irreversibility/disruption costs.

5 Estimation strategy

5.1 Indirect inference

The structural parameter vector is estimated by minimizing the distance between simulated model moments and data moments:

$$\hat{\Theta} = \arg \min_{\Theta} [m^{data} - m^{model}(\Theta)]' W [m^{data} - m^{model}(\Theta)],$$

where m^{data} contains the empirical moments and $m^{model}(\Theta)$ contains the simulated moments generated by a candidate parameter vector.

5.2 Why indirect inference is useful here

- The structural model has non-linear and occasionally non-smooth policy rules.
- Direct likelihood estimation is difficult because plant-level investment has lumpy, intermittent, and skewed behavior.
- Moment matching allows the authors to focus on the empirical features most informative about adjustment costs.

6 Tables and figures

6.1 Figure 1 and Table 1: investment-rate distribution and summary statistics

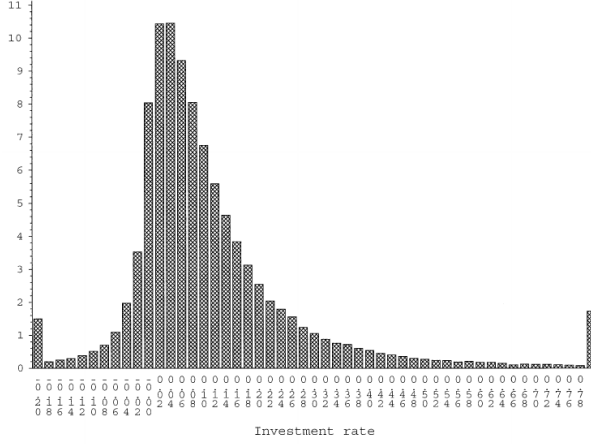


FIGURE 1
Investment rate distribution

TABLE 1
Summary statistics

Variable	LRD
Average investment rate	12.2% (0.10)
Inaction rate: investment	8.1% (0.08)
Fraction of observations with negative investment	10.4% (0.09)
Spike rate: positive investment	18.6% (0.12)
Spike rate: negative investment	1.8% (0.04)
Serial correlation of investment rates	0.058 (0.003)

TABLE 2
Parameterization of illustrative models

Model	γ	F	λ	p_s	p_b
No AC	0	0	1	1	1
CON	2	0	1	1	1
NC-F	0	0.01	1	1	1
NC- λ	0	0	0.95	1	1
TRAN	0	0	1	0.75	1

TABLE 3
Moments from illustrative models

Moment	LRD	No AC	CON	NC-F	NC- λ	TRAN
Fraction of inaction	0.081	0.0	0.038	0.616	0.588	0.69
Fraction with positive investment bursts	0.18	0.298	0.075	0.212	0.213	0.120
Fraction with negative investment bursts	0.018	0.203	0.0	0.172	0.198	0.024
Corr (i_{it}, d_{it-1})	0.058	-0.053	0.732	-0.057	-0.06	0.110
Corr (i_{it}, a_{it})	0.143	0.202	0.692	0.184	0.196	0.346

Tables 2 and 3: illustrative model parameterizations and simulated moments.

Figure 1 and Table 1: investment-rate distribution and LRD summary statistics.

Read:

- Figure 1 shows a skewed investment-rate distribution with many observations near low investment and a long right tail of positive investment spikes.
- Table 1 reports the key LRD moments: average investment is 12.2%, inaction is 8.1%, negative investment is 10.4%, positive spike rate is 18.6%, negative spike rate is 1.8%, serial correlation is 0.058, and the correlation between profit shocks and investment is 0.143.
- Tables 2 and 3 compare illustrative models: no adjustment costs, convex costs, non-convex fixed costs, irreversibility/resale frictions, and disruption costs.

Economic message: No simple model matches all the moments. The data require both lumpy adjustment and a muted, non-excessively-persistent response to shocks.

6.2 Table 4: estimates with $\lambda = 1$

TABLE 4
Parameter estimates: $\lambda = 1$

Spec.	Structural parameter estimates (S.E.)			Moments				
	γ	F	p_s	Corr (i, i_{-1})	Corr (i, a)	Spike ⁺	Spike ⁻	$\mathcal{L}(\hat{\theta})$
LRD								
all	0.049 (0.002)	0.039 (0.001)	0.975 (0.004)	0.058 0.086	0.143 0.31	0.186 0.127	0.018 0.030	6399.9
γ only	0.455 (0.002)	0	1	0.605	0.540	0.23	0.028	53,182.6
p_s only	0	0	0.795 (0.002)	0.113	0.338	0.132	0.033	7673.68
F only	0	0.0695 (0.00046)	1	-0.004	0.213	0.105	0.0325	7390.84

LRD, Longitudinal Research Database.

Table 4: parameter estimates with $\lambda = 1$.

TABLE 5
Parameter estimates: $F = 0$

Specification	Structural parameter estimates (S.E.)			Moments				
	γ	λ	p_s	Corr (i, i_{-1})	Corr (i, a)	Spike ⁺	Spike ⁻	$\mathcal{L}(\hat{\theta})$
LRD								
λ only	0	0.796 (0.0040)	1.0	0.058 -0.009	0.143 0.06	0.186 0.107	0.018 0.042	9384.06
All	0.153 (0.0056)	0.796 (0.0090)	0.981 (0.0090)	0.148	0.156	0.132	0.023	2730.97

LRD, Longitudinal Research Database.

Table 5: parameter estimates with $F = 0$.

Read:

- Table 4 estimates models without disruption costs ($\lambda = 1$). The full specification estimates a small convex cost, a fixed cost, and a resale parameter below one.
- The model with only quadratic/convex costs performs poorly because it generates too much serial correlation and too strong a contemporaneous investment-profitability correlation.
- The model with fixed costs only or irreversibility only also misses important moments.

Economic message: Even when disruption costs are shut down, the estimates point toward a mixture of convex and non-convex frictions. However, the model fit leaves room for improvement.

6.3 Table 5: estimates with disruption costs

Read:

- Table 5 allows disruption costs through $\lambda < 1$ while setting $F = 0$.
- The row labelled “All” estimates $\gamma = 0.153$, $\lambda = 0.796$, and $p_s = 0.981$.
- The model fits much better than the fixed-cost specification in Table 4, as shown by the lower objective statistic.
- The disruption-cost-only model improves some features but still does not fit as well as the mixed specification.

Economic message: Disruption costs are important. Capital adjustment appears to impose production-disruption losses in addition to smooth convex costs and irreversibility-like resale frictions.

6.4 Table 6: sectoral estimates

TABLE 6
Sectoral parameter estimates

Sector	Parameter estimates (S.E.)				Moments				
	γ	F	λ	p_s	$\text{corr}(i, i_{-1})$	$\text{corr}(i, a)$	spike^+	spike^-	$\mathbb{E}(\hat{\theta})$
331-LRD					0.086	0.133	0.064	0.02	n/a
331-est.	0.0 (0.0003)	0.07 (0.0017)	1	0.946 (0.0046)	-0.03	0.249	0.076	0.022	62.19
	0.015 (0.0037)	0	0.70 (0.0344)	0.76 (0.0167)	0.041	0.146	0.073	0.018	8.13
371-LRD					0.082	0.132	0.204	0.013	n/a
371-est.	0.012 (0.0026)	0.069 (0.0057)	1	0.962 (0.0106)	0.069	0.338	0.112	0.037	452.96
371-est.	0.051 (0.0068)	0	0.679 (0.0398)	0.8082 (0.0218)	0.251	0.132	0.096	0.027	318.04

LRD, Longitudinal Research Database; est., estimates.

Table 6: sectoral parameter estimates.

Read:

- Table 6 re-estimates the model for selected sectors, including steel and transportation.
- The sectoral evidence again supports both convex and non-convex adjustment costs.
- The specification with disruption costs often performs better than the pure fixed-cost specification.

Economic message: The main conclusion is not only an aggregate artifact. The evidence for multiple adjustment frictions also appears in sectoral exercises.

7 Short summary

- The paper studies plant-level capital adjustment using the LRD.
- The investment-rate distribution is skewed, lumpy, and non-normal.
- The authors estimate structural adjustment-cost parameters using indirect inference.
- Pure quadratic adjustment costs cannot match the low serial correlation and burstiness of investment.
- Pure non-convex models also fail to match all moments.
- The best-fitting model mixes convex and non-convex costs, especially disruption costs and irreversibility-like frictions.
- The results imply that capital adjustment is not governed by one simple friction; it reflects multiple margins of adjustment.

8 Conclusion

8.1 Core conclusion

Cooper and Haltiwanger show that plant-level investment behavior is best explained by a combination of convex and non-convex adjustment costs. The model that mixes smooth convex costs with disruption costs and irreversibility-like frictions matches key investment moments more successfully than models with only one adjustment-cost component.

8.2 Economic intuition

Plants do not adjust capital continuously and smoothly in response to every shock. Adjustment disrupts production, resale is imperfect, and some changes are costly enough to delay until the benefits justify a larger investment burst. At the same time, investment is not governed only by fixed costs: the intensive margin still matters, and convex costs help explain gradual responsiveness to profitability.

8.3 Incremental contribution revisited

The paper advances the investment literature by moving from documenting lumpy investment to structurally estimating the costs that generate it. It also shows that matching plant-level investment moments requires a richer cost structure than the canonical quadratic model.

8.4 Limitations

The estimates depend on the chosen moments, the structural specification, and the simulated profitability process. The model is deliberately stylized, so specific estimated cost parameters should be interpreted through the lens of the model. Still, the qualitative result - that multiple adjustment frictions are needed - is robust across several specifications.

8.5 Takeaway

The enduring takeaway is that capital adjustment costs are heterogeneous and multidimensional. For empirical investment work and policy counterfactuals, assuming a single smooth quadratic adjustment cost can miss the central microeconomic features of plant-level capital adjustment.